



'Print-to-Release' Regional Specific Human iPSc-Derived Spinal Assembloids for Chronic Spinal cord Injury Repair

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Background

- **Spinal Cord Injury (SCI) and Deficits:** SCI leads to sensory and motor deficits resulting from damage to the dorsal and ventral spinal cord, respectively.
- **Importance of Regional Specification:** Patterning grafted cells to match the phenotypic identity of the injury site enhances integration.
- **Cell Types and Functions:**
 - Ventral spinal neural progenitor cells (vsNPCs): Support motor circuit function.
 - Dorsal spinal neural progenitor cells (dsNPCs): Aid in sensory relay.
- **Integration:** Achieving correct regional specification is essential for transplanted cells to integrate effectively with the native spinal cord.

Key Hurdles in Spinal Cord Regenerative Medicine

- **Cellular Integration Failure:** Dissociated cells often lack proper engraftment, resulting in poor tissue integration or cell death.
- **Lack of Regional Identity:** Unspecified progenitors frequently fail to develop the precise regional identity necessary for neural circuit reconstruction.
- **Efficacy and Immunological Limitations:** Current clinical trials using fetal/embryonic sources show limited success and require immunosuppression.
- **Regulatory Complexity:** Combining cell therapies with scaffolds creates significant FDA regulatory hurdles.
- **Logistical and Economic Feasibility:** Autologous iPSC therapies remain cost-prohibitive and practically unsustainable.

Established Data from Our Lab

Establishment of dorsal and ventral spinal neural progenitor cells



Fig 1. Directed differentiation of dorsal and ventral spinal neurons. Following an accelerated 24-hour neural1 induction, cells may be dorsalized or ventralized (data not shown) to neural progenitors at day 6 or post-mitotic at day 21. [1-3]

3D printed organoid scaffolds transplanted into a complete transection SCI at T7/T8 athymic nude (ATN) rats

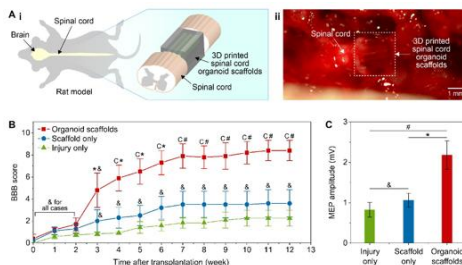


Fig 2. 3D-printed organoid scaffolds A. bridging T8/9 transection in athymic nude rats. B. BBB scores improve significantly from weeks 3–12 in the organoid scaffold group vs. Controls (repeated-measures ANOVA; *p<0.05, #p<0.01, #p<0.001, *p=NS). C. MEP amplitudes significantly higher at week 12, indicating restored motor conduction across the lesion (one-way ANOVA + Bonferroni; *p<0.05, #p<0.01, &p=NS). Mean ± SEM.

A Novel Scaffold-Free Organoid Transplantation Method

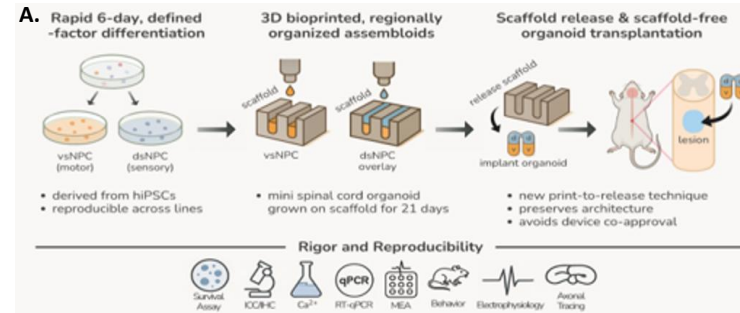
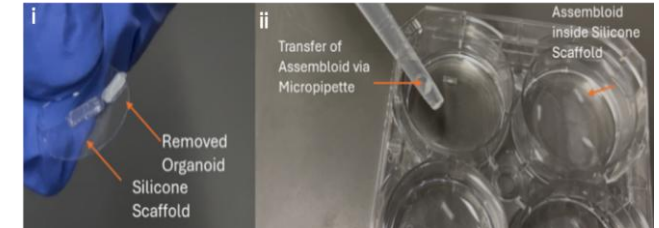


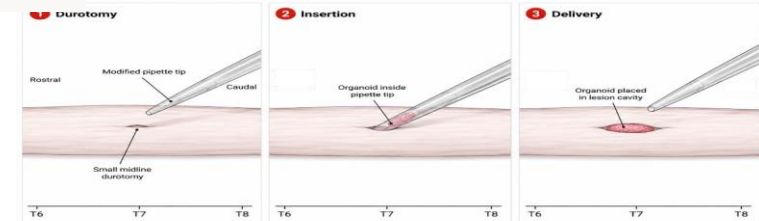
Fig 3. Print-to-Release technique

Human iPSC derived dsNPCs and vsNPCs are 3D bioprinted into a silicone scaffold with organized dorsal-ventral architecture and cultured for 21 days. A. Novel print-to-release technique, B. The assembloid is detached from the scaffolds and C. Implantation of scaffold free assembloids into the spinal cord. This approach preserves cellular architecture, avoids biomaterial co-approval, and enables rigorous pre- and post-implant evaluation.

B. Preparation of assembloids for transplantation



C. Preparation of assembloids for transplantation



Proof of Concept - Results

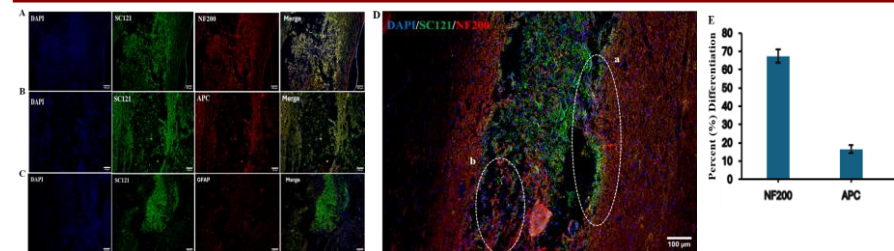


Figure 4. Differentiation pattern of human iPSC-derived organoids in vivo. Representative images of DAPI-labeled cells with neural markers: A. NF200, B. APC, and C. GFAP. D. Human SC121⁺ neurons (a) integrated into the injury site and (b) formed connections with host NF200⁺ neurons. E. Percentage of differentiation of the cells. SC121⁺ neurons made up the majority of the cells, whereas a smaller fraction became oligodendrocytes and astrocytes.

Summary and Future Directions

- Our lab utilized a combinatorial approach to restore anatomical and functional outcomes in chronically transected rats by transplanting a stacked, sandwiched 3D bio-scaffold containing regionally specified upper dorsal (DsNPC) and lower ventral (VsNPC) organoids.
- Future studies will evaluate scaffold-free "assembloids" (fused dorsal and ventral organoids) in a contusion model of chronic spinal cord injury in rats
- Positive preclinical results will inform IND-enabling studies using CIRa's cGMP master bank, targeting first-in-human evaluation of scaffold-free iPSC-derived spinal assembloids in chronic SCI.

References

1. Huntemer-Silveira A et al. Accelerated differentiation of hiPSCs into regionally specific spinal NPCs. *Front Neurosci.* 2023;17:1251906.
2. Walsh P et al. Defined culture conditions accelerate small-molecule-assisted neural induction. *Cell Transplant.* 2017;26(12):1890.
3. Han G et al. 3D-Printed Scaffolds Promote Enhanced Spinal Organoid Formation. *Adv Healthc Mater.* 2025;14(24):e04817.

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